

v SAVING LIVES
Adar Poonawalla at the SII headquarters in Pune

cally neutralised), a person's immune system will respond and develop antibodies against it. So, when the person is exposed to the actual virus, the antibodies he had developed would attack the invader and neutralise it.

Though more taxing and cumbersome, Ella chose the killed or inactivated virus route as he had already developed vaccines against rabies, diarrhoea and Japanese encephalitis using the same technique. He was also aware that given the high stakes, failure, especially on safety norms, could lead to mass litigation and criminal action against his firm. So, instead of developing the vaccination on his own, he sought collaboration with a central government-owned entity. Ella homed in on the National Institute of Virology (NIV), a reputed research institution that came under the Indian Council of Medical Research (ICMR). His reason: "We were working with a live vaccine that was highly contagious and it was dangerous for even our employees to handle it. I wanted to partner with the government because we could do everything transparently. NIV had the kind of sophisticated facilities to conduct tests on vaccines for safety, efficacy and dosage. I saw value in this public-private partnership that could give India a global presence."

At NIV, Dr Priya Abraham, its feisty director, was waiting and ready. In December 2019, when word got out that a new kind of pneumonia had broken out in Wuhan in China, Abraham says, "Our warning antenna was already bobbing." Their concern was that the virus could soon come to India and, by the first week of January, the NIV's National Influenza Centre began acquiring the primers needed to do the reverse-transcript polymerase chain reaction (RT-PCR) test to detect the virus. On January 29, when Abraham was on a trip to Delhi, she received a call from her colleague in Pune that they had detected the first positive case in India. Abraham rushed back and decided to get one of their other laboratories to repeat the test on the sample to confirm the finding.

When Abraham subsequently saw an image of Covid-19, she remarked, "It looked like a pretty virus. It had spikes on it that stuck out like the rays of a sun or like the protrusion of a crown prettily arranged...in looks, it was a classic Corona virus." Since NIV had a BSL-4 facility, they were among the first institutes in the world to isolate the alpha variant of the virus, decipher its genome structure and subsequently grow it in vitro at its facility. That was when Ella made a formal request to ICMR director-general Dr Balram Bhargava to permit NIV to collaborate with BBIL in the development of the indigenous vaccine.

Meanwhile, Bhargava and his team had already surveyed potential vaccine candidates across the world. Multinational pharmaceutical companies like Moderna, Pfizer and Johnson & Johnson were in an advanced stage of making vaccines using newer gene technology but, apart from limited production, many of these vaccines required high-end cold chains of minus 70 degrees that would strain India's health infrastructure.